Cambridge (CIE) A Level Business

Your notes

Marketing Analysis (A Level)

Contents

- Price Elasticity of Demand
- Income Elasticity of Demand
- Promotional Elasticity of Demand
- Product Development
- Sales Forecasting

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The concept of price elasticity of demand

- **Price elasticity of demand (PED)** measures how **responsive** demand for a product is to a change in **price**
  - In general,
    - when the price for a product increases, demand falls
    - when the price for a product decreases, demand rises
  - PED answers the question: **By how much** will demand change?

Calculating price elasticity of demand

- PED will always be a **negative value** due to the **inverse relationship** between price and quantity
  - If the price goes up, the quantity demanded goes down
  - If the price goes down, the quantity demanded goes up
- **PED can be calculated** using the formula

$$\text{PED} = \frac{\text{\% Change in quantity demanded}}{\text{\% Change in price}} = \frac{\%\,\Delta \text{ in QD}}{\%\,\Delta \text{ in P}}$$

- **To calculate a % change**, use the formula

$$\text{\% Change} = \frac{\text{New value} - \text{Old value}}{\text{Old value}} \times 100$$

Worked Example

Helsinki Hurricanes sells team shirts for $24. In 2024 it sold 3,240 shirts at this price. Due to increased costs, the team manager has decided to increase the price to $30 per shirt from 2025. She expects demand to fall by 12%.

Calculate the price elasticity of demand for Helsinki Hurricanes team shirts.

(2)

Step 1: Calculate the percentage change in price

$$= \frac{\$30 - \$24}{\$24} \times 100 \quad [1]$$

$$= 25\%$$

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Step 2: Divide the percentage change in demand by the percentage change in price

$$= \frac{-12\%}{25\%} \quad [1]$$

$$= -0.48$$

- In some cases, you may be required to calculate the percentage change in price or demand
- This involves rearranging the PED formula

Worked Example

The price elasticity of demand for popcorn at the cinema is –0.8. The current price of a box of popcorn is $5.

Using the data, **calculate** the percentage change in quantity demanded following a $1 increase in the price of a box of popcorn. You are advised to show your work.

[4]

Step 1: State the PED formula

$$\text{PED} = \frac{\text{\% Change in demand}}{\text{\% Change in price}} \quad (1)$$

Step 2: Calculate the percentage change in price

$$= \frac{\$6 - \$5}{\$5} \times 100 \quad (1)$$

$$= 20\% \text{ increase}$$

Step 3: Insert the data you have been given into the formula

$$-0.8 = \frac{x}{20\%} \quad (1)$$

Step 4: Rearrange and solve for x

$$x = -0.8 \times 20 \quad (1)$$

$$x = -16\%$$

Step 5: Present the final answer

- The quantity demanded **falls** by **16% (4)**

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Examiner Tips and Tricks

Remember the key facts:

- If price decreases, demand increases
- If price increases, demand decreases

In the example above, price increases, so demand must fall

Interpreting price elasticity of demand

1. Price elastic demand

- PED value lower than -1 (e.g -1.2)
- Demand is **more responsive to a change in price**
  - For every 1% change in price, demand will change by more than 1%
- An increase in price will lead to a fall in revenue; a decrease in price will lead to an increase in revenue
  - Examples include **luxury products** such as cars, smartwatches, foreign holidays, cinema visits, jewellery and branded goods

2. Price inelastic demand

- PED value is between 0 and -1 (e.g -0.7)
- Demand is **less responsive to a change in price**
  - For every 1% change in price, demand will change by less than 1%
- An increase in price will lead to an increase in revenue; a decrease in price will lead to an decrease in revenue
  - Examples include **necessities** such as bread, milk, eggs and potatoes; fuel; rent; toothpaste
    - Also **addictive products** such as cigarettes and sugary foods

Examiner Tips and Tricks

Focus on the **size** of the number

–1.2 is price elastic, –0.5 is price inelastic. Always link what that means for the business revenue when the price changes

The impact of price elasticity of demand on pricing decisions

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- If businesses can **determine the price elasticity of demand** for their products, they can adjust **pricing strategy** to **maximise revenue**
- If demand for their products is relatively **price inelastic (between 0 and –1),** raising the price will lead to an **increase in total revenue**
- However, lowering the price will lead to a **fall in total revenue**
  - **Price skimming strategies** are best employed for products that are price inelastic in demand
- If demand for their products is relatively **price elastic (PED > –1),** raising the price will lead to a **fall in total revenue**
- However, lowering the price will lead to a **rise in total revenue**
  - **Competitive pricing strategies** are best employed for products that are price elastic in demand

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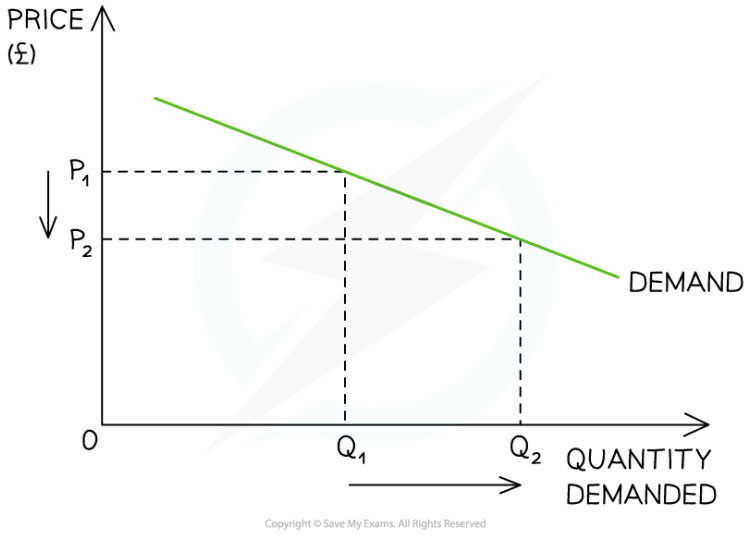
Price elasticity of demand and total revenue

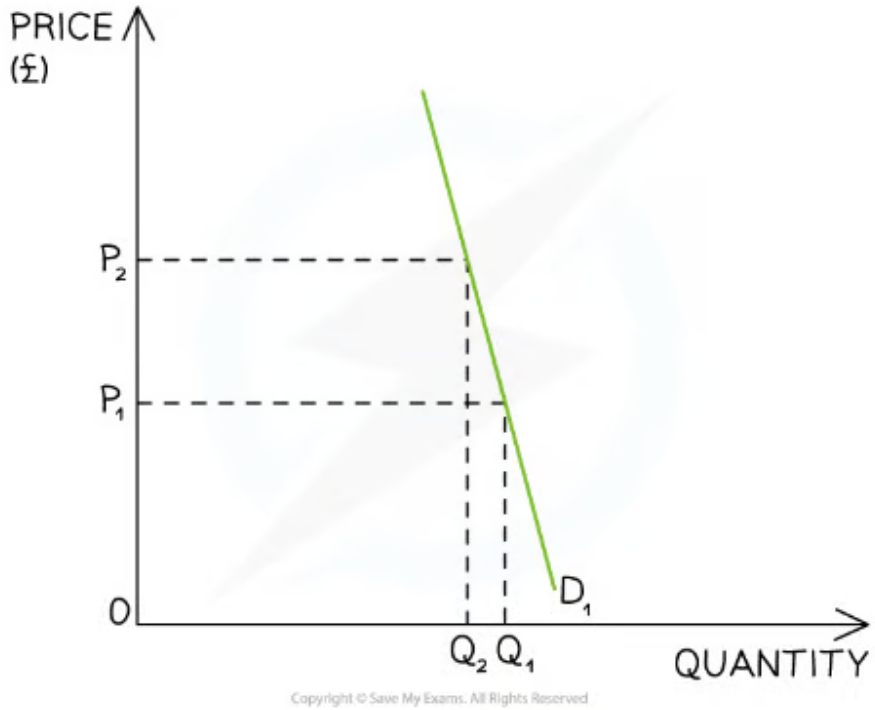
Price elastic demand



Price elastic demand

- PED is **greater than 1**
- An increase in selling price reduces the total amount of revenue generated from sales
- A reduction in selling price increases the total amount of revenue generated from sales

Price inelastic demand



Price inelastic demand

- PED is **between 0 and 1**
- An increase in selling price increases the total amount of revenue generated from sales
- A reduction in selling price reduces the total amount of revenue generated from sales

Limitations of price elasticity of demand

- **Price elasticity of demand (PED)** can help businesses decide how much to charge for their products by showing how sensitive customers are to price changes
- However, relying too much on PED when making decisions can lead to problems, especially if the data is wrong or if other important factors are ignored

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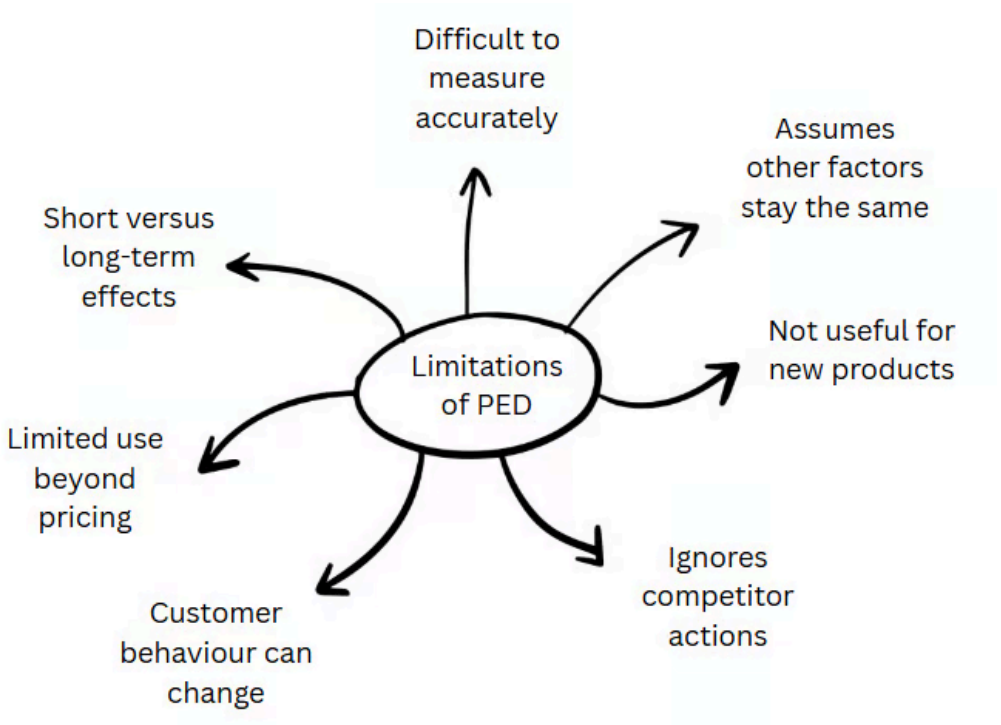

Limitations of PED include the difficulty of measuring demand change accurately and its limited use beyond pricing

- **Difficult to measure accurately**
  - PED calculations need reliable data on how demand changes with price, which can be hard to collect
    - E.g. A business may not know exactly how much demand will fall if it raises prices by 10%
- **Assumes other factors stay the same**
  - PED only looks at price and demand, but things like income, trends or marketing also affect sales
    - E.g. A fashion brand may raise prices expecting little change in demand, but a trend shift makes the item less popular
- **Not useful for new products**
  - Without past data, it's hard to estimate PED for products that haven't been sold before
    - E.g. A tech company launching a new gadget can't predict how price changes will affect demand
- **Ignores competitor actions**
  - PED doesn't consider how rival businesses might respond to a price change
    - E.g. A supermarket lowers prices to boost sales, but rivals do the same, so demand stays unchanged

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- **Customer behaviour can change**
  - PED is not fixed – it can change as customer preferences or incomes change
    - E.g. Petrol demand may become more elastic as electric cars become more common
- **Limited use beyond pricing**
  - PED mainly helps with pricing decisions, not choices like product design, location or hiring
  - However, it can give clues about brand strength or customer loyalty, which may influence other areas
- **Short versus long-term effects**
  - Demand might be inelastic in the short term but more elastic in the long term as customers adjust
  - E.g. A mobile network raises prices and sees little change at first, but over time people switch providers

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The concept of income elasticity of demand

- **Income elasticity of demand (YED)** measures the **responsiveness** of demand for a product to a change in **income levels**
  - Businesses are interested **in how much** the **quantity demanded will change** for different products
- When household incomes change, spending on goods and services will **either increase or decrease,** depending on the type of good or service it is
  - **Normal good:** A good where demand rises when income increases, e.g. restaurant meals or new clothes
  - **Inferior good:** A good where demand falls when income increases, e.g. instant noodles or bus travel
  - **Luxury good:** A good where demand increases more than proportionally as income increases, e.g. designer handbags or sports cars

Calculating and interpreting income elasticity of demand

- The YED value can be **positive or negative,** and the value is important in determining the type of good
  - A good with a **positive YED** value is considered to be a **normal good**
    - Normal goods can be classified as **necessities** or **luxuries**
  - A good with a **negative YED** value is considered to be an **inferior good**
- YED is calculated using the formula

$$\text{YED} = \frac{\text{\% Change in quantity demanded}}{\text{\% Change in income}} = \frac{\%\Delta \text{ in QD}}{\%\Delta \text{in Y}}$$

Worked Example

An individual's income falls from £450 to £405 per week. As a result, their demand for takeaway meals falls from 50 to 30 per week.

Calculate the income elasticity of demand (YED) for takeaway meals.

[4]

Step 1: State the YED formula

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$$\text{YED} = \frac{\text{\% change in QD}}{\text{\% change in Y}} \quad (1)$$

Step 1: Calculate the % change in QD

$$\text{\% change in QD} = \frac{50 - 30}{50} \times 100 \quad (1)$$

$$= -40\%$$

Step 2: Calculate the % change in Y

$$\text{\% change in Y} = \frac{£450 - £405}{£450} \quad (1)$$

$$= -10\%$$

Step 3: Insert the above values in the YED formula

$$\text{YED} = \frac{-40\%}{-10\%} \quad (1)$$

$$= 4$$

Influences on YED

- YED is influenced by **many factors in an economy** that **change the wages of workers**
  - During a **recession,** wages usually fall and demand for **inferior goods** rises while demand for **luxury goods** falls
  - During a period of **economic growth** and rising wages, demand for luxury goods increases while demand for inferior goods decreases
  - Other influences on income include **minimum wage legislation**, **taxation** and **increased international trade**
- Understanding the income elasticity of demand is useful to businesses, as it can **help** them **plan their production and products**
  - Planning in this way will help them generate higher profits and be less exposed to economic downturns

Interpreting YED values

YED is more than 1 and positive

- **Normal good**
  - Examples: cars, foreign holidays, cinema visits and branded goods
  - Demand rises when income rises

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- Demand falls when income falls
- Demand is responsive to a change in income (income elastic)

YED is between 0 and 1 and positive

- **Necessity good**
  - Examples: bread, milk, fuel, toothpaste
  - Demand rises slightly as income rises
  - Demand falls slightly as incomes fall
  - Demand is not very responsive to a change in income (income inelastic)

YED is negative

- **Inferior good**
  - Examples: public transport, domestic holidays, canned foods, own-label goods
  - Demand rises when income falls
  - Demand falls when income rises
  - Negative income elasticity

The impact of income elasticity of demand on decisions

Pricing and promotion

- **Firms can anticipate changes in demand** when incomes rise or fall:
  - Products with **high positive YED** (luxury goods) will see demand increase more than proportionally as income rises
  - During a downturn, demand for such products may **fall sharply**, so businesses may need to **lower prices or offer discounts**
  - For **necessities** (low YED, close to zero), demand is **less sensitive to income**, so businesses can maintain prices even in economic downturns

Product and capacity planning

- **Positive income elasticity** signals a non-necessity or luxury item, where sales rise as incomes rise
  - **Burberry** enjoys strong growth when wages improve, so the firm expands its high-end ranges and store space
- **Negative income elasticity** identifies inferior goods
  - Supermarket "value" noodles sold more during the 2022–23 cost-of-living squeeze, so factories increased night shifts to cope

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- **Oatly**'s premium oat milk drink saw fast UK sales growth as incomes recovered in 2024, so the brand invested in a new Peterborough packing facility

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Marketing objectives

- Firms set objectives using elasticity estimates
  - **B&Q** identified that spending on DIY products is income elastic, so it continued to add products to its portfolio during the recent economic downturn

Inventory and production schedules

- Knowing that Pumpkin Spice Lattes have a high positive PED but a strong seasonal YED, **Starbucks** launches them every August and adjusts its coffee bean orders months ahead

Risk management

- Airlines such as **Ryanair** use dynamic pricing models built on PED estimates
  - If a flight is 80% full, prices rise sharply because late bookers are less price-sensitive

Examiner Tips and Tricks

If you're using PED or YED data, don't stop at calculating it

Always say how it affects pricing or sales decisions

Limitations of income elasticity of demand

- Income elasticity of demand helps businesses understand if their product is a luxury, necessity or inferior good
- However, just like with PED, there are limits to how useful YED is when making business decisions

Key limitations of YED

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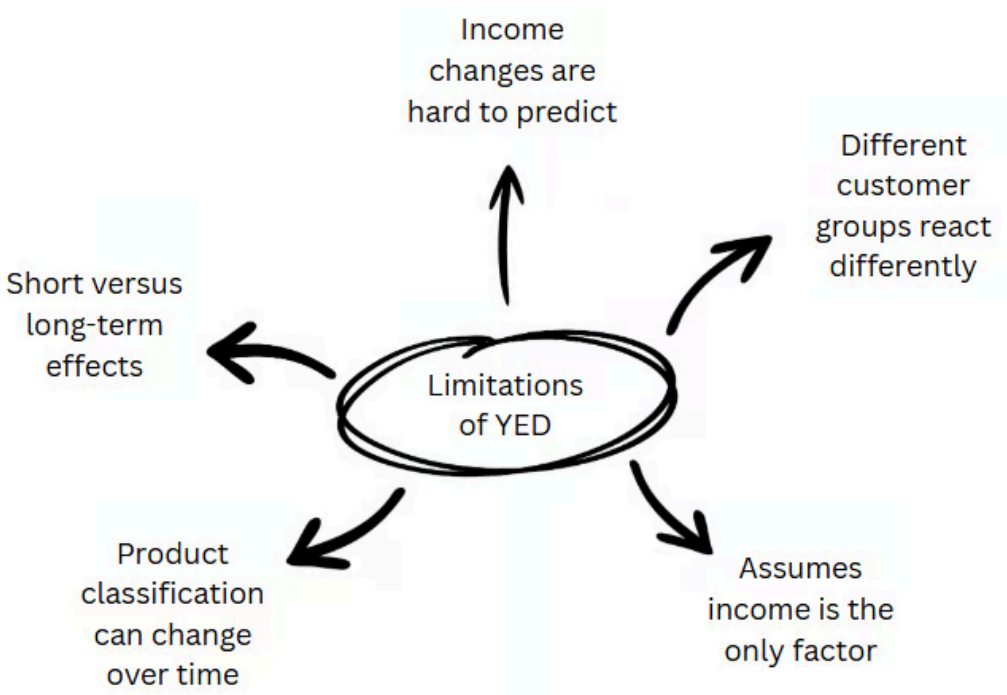

Limitations of YED include the different reactions of customer groups and the assumption that income is the only factor affecting demand

- **Income changes are hard to predict**
  - It's difficult to know exactly when or how much people's income will rise or fall
    - E.g. A sudden cost-of-living crisis could reduce incomes and demand for luxury items faster than expected
- **Different customer groups react differently**
  - Not everyone's income changes in the same way, so demand might not shift evenly
    - E.g. A business may target high earners, but average YED data includes lower-income groups too
- **Assumes income is the only factor**
  - YED focuses on income changes but ignores other factors like trends, advertising or substitutes
    - E.g. A healthy food company may see demand rise due to a trend, not just because incomes have increased
- **Product classification can change over time**
  - A product that is seen as a luxury today might become a necessity tomorrow
    - E.g. Smartphones were once luxuries, but now they are widely considered to be essential
- **Short-term versus long-term effects**

- Demand might react slowly to income changes, especially if customers delay spending
    - E.g. After getting a pay rise, an individual may not upgrade their car straight away

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The concept of promotional elasticity of demand

- Promotional elasticity of demand measures the **responsiveness of demand** for a product to a change in the **amount of money spent on promoting it**
- In most cases, an **increase in spending on promotional activity should increase demand** for a product, whilst a **reduction in spending on promotional activity should lead to a fall in demand**
  - Promotional elasticity of demand determines **by how much** the level of demand will change when spending on promotional spending is increased or decreased
- Businesses can **use promotional elasticity** of demand to determine
  - Whether increased spending on promotion is **worthwhile**
  - The **effect on demand** of reducing spending on promotional activity
  - The **best use** of promotional budgets

Calculating promotional elasticity of demand

- Promotional elasticity is calculated using the formula:

$$\text{Promotional elasticity of demand} = \frac{\text{\% change in quantity demanded}}{\text{\% change in promotional spending}} = ? : 1$$

- The outcome is expressed as a **ratio**
  - This shows the **percentage by which demand changes with each one per cent change in promotional spending**

Worked Example

Yaltrex Industries currently spends $2.4 million each year promoting its range of e-scooters and e-bikes. It is considering running a TV advertising campaign costing $600,000 that it expects will increase annual sales by 30%.

Calculate the promotional elasticity of demand for Yaltrex's proposed TV advertising campaign.

[3]

Step 1: Calculate the percentage change in promotional spending

$$\text{Percentage change} = \frac{\text{Change in spending}}{\text{Original spending}} \times 100$$

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$$= \frac{600,000}{2,400,000} \times 100$$

$$= 25\% \quad [1]$$

Step 2: Calculate promotional elasticity of demand

$$\text{Promotional elasticity of demand} = \frac{\%\ \text{change in quantity demanded}}{\%\ \text{change in promotional spending}}$$

$$= \frac{30\%}{25\%} \quad [1]$$

$$= 1.2 : 1 \quad [1]$$

- **In this case, for every 1% increase in promotional spending, demand is expected to rise by 1.2%**

Interpreting and using promotional elasticity to make decisions

- Businesses can use promotional elasticity of demand to determine **whether changes in promotional spending are worthwhile**
  - Where **demand is responsive (elastic)** to changes in promotional spending, **increasing or decreasing spending** can have a **significant impact on product sales**
    - The percentage change in demand will be greater than the percentage change in promotional spending
    - The outcome of the formula will be **greater than 1**
    - In these cases, increased spending on promotional activity **could be worthwhile**
    - **Reducing spending** on promotional activity could cause a **significant drop in demand** so may be best avoided
  - Where **demand is unresponsive (inelastic)** to changes in promotional spending, **increasing or decreasing spending** will have a **limited impact on product sales**
    - The percentage change in demand will be less than the percentage change in promotional spending
    - The outcome of the formula will be **less than 1**
    - In these cases, increased spending on promotional activity **is unlikely to be worthwhile**
    - **Reducing spending** on promotional activity is likely to have **little impact on demand** so may be an effective way to reduce costs

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Worked Example

Vulkan Tours Ltd's marketing director has proposed increasing advertising spending in 2024 by 55% to pay for monthly magazine advertisements. She hopes these adverts will increase sales of its specialist educational travel packages by 25% over the year.

Explain why Vulkan Tours Ltd's directors may reject this advertising proposal.

[3]

Step 1: Calculate the promotional elasticity of demand

$$\text{Promotional elasticity of demand} = \frac{\text{\% change in quantity demanded}}{\text{\% change in promotional spending}}$$

$$= \frac{25\%}{55\%}$$

$$= 0.45 : 1 \quad \textbf{[1]}$$

Step 2: Explain the outcome

- Vulkan Tours Ltd's directors may reject this proposal because **demand in response to changes in promotional spending is inelastic [1]**
- For every **1% increase in promotional spending, demand is expected to rise by just 0.45% [1]**, indicating that this significant increase in spending may not be worthwhile

- Promotional elasticity of demand can also help managers **evaluate promotional options** and select the option that is **most likely to increase demand**
  - The **higher the outcome** of the formula, the **more responsive** demand is to changes in promotional spending
    - **Focusing spending** on products with the **highest promotional elasticity of demand** is likely to be the **best use of the promotional budget,** as it is most likely to **lead to increased sales**
  - However, consideration of the **affordability** of promotional activity is also important

Worked Example

Flatfish Scuba's marketing manager is trying to decide how to promote its new range of watersports equipment. She has determined the cost of three options and estimated the promotional elasticity of demand for each.

Option	Cost ($)	Promotional Elasticity of Demand

1. Advertising in specialist magazines	22,000	1.4 : 1
2. A viral campaign using social media	18,000	1.3 : 1
3. Sponsoring a national aquatic event.	40,000	1.6 : 1

Explain **one** benefit and **one** drawback of exclusively using promotional elasticity of demand to determine which option the marketing manager should choose. **[6 marks]**

Step 1: Explain one benefit

- Promotional elasticity of demand shows how much demand is expected to increase with a change in promotional spending **[1]**
- In this case, the marketing manager can see that sponsoring a national aquatic event would likely have the greatest impact on demand **[1]** because it has the highest value
- For every $1 spent on this promotional activity, demand for Flatfish Scuba's products would increase by 1.6% **[1]**.

Step 2: Explain one drawback

- However, exclusively using promotional elasticity of demand to determine the choice of promotional activity does not take into account the cost of the promotion **[1]**
- In this case, sponsoring a national aquatic event is the most expensive option, around twice the cost of the other options **[1]**
- Flatfish Scuba could spend the same amount on both advertising in specialist magazines and a viral social media campaign and achieve better results **[1]**

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Limitations of promotional elasticity of demand

- **Promotional spending doesn't always aim to boost sales straight away**
  - Some promotions are used to build a stronger brand or increase customer loyalty, which may not show in short-term demand changes
- **Promotional elasticity doesn't show the full picture**
  - It only measures the link between spending and sales, not how the promotion affects the brand in the long term
- **Hard to measure the impact of one promotion**
  - When a business promotes lots of products at once, it's tricky to see how much impact one campaign had on demand
- **Other things can affect demand during a promotion**
  - Changes in taste, word-of-mouth, or competitor actions can all influence sales while a promotion is running
- **Promotions often run over a long time**

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- It is difficult to link demand changes to a specific time, especially if the promotion has no clear start or end date

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Examiner Tips and Tricks

Elasticity of demand calculations often require the use of estimates. Questioning the accuracy of these estimates is a good evaluative point, especially of calculations are relied upon to make business decisions.

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The process of product development

- One way to stay ahead of the competition is by **developing new products** and innovating **existing ones**
- New product development can be **valuable** for a number of reasons
  - A successful launch opens up a **new income stream**
    - **Nintendo**'s launch of its Switch console in 2017 turned falling sales into record profits
  - **Tastes change**, so fresh products keep a business in fashion
    - **Skechers Slip Ins** are easy to put on, making them a comfortable alternative to lace-up shoes that align well with less formal working lives
  - Novel features create a **competitive advantage** over rivals
    - The **Samsung Galaxy Z Fold** offers a folding screen that mainstream phones lack
  - Extra product lines **spread risk** across different segments
    - The **LEGO Botanical Collection** targets adult hobbyists, not just children, diversifying sales
  - Regular innovation can improve brand image
    - **Unilever's Dove** refillable deodorant pods underline its sustainability credentials

Key factors that influence new product development

Factor	Why it shapes the new product plan	Example
Market demand and insight	▪ Real customer problems or gaps in the market can prompt profitable ideas	▪ The trend for flexitarian diets led **Burger King** to create the plant-based Whopper
Technology advances	▪ New technology unlocks features that were impossible before	▪ Wider 5G networks encouraged **Qualcomm** to design faster chips for gaming consoles and computers
Competitor activity	▪ A rival's move can force a business to respond rapidly	▪ **Microsoft** added haptic feedback to its Elite gaming controllers after Sony's success with its DualSense product

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Resources and R&D budget	▪ Innovation requires time, talent and cash	▪ **Pfizer** channelled major R&D funds into mRNA tech, enabling it to develop its COVID-19 vaccine quickly

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The product development process

- There are several steps in the product development process

The new product development process

1. **Generate ideas**
   - Product concepts are discussed and brainstormed using customer suggestions, ideas from competitors' products, employees' ideas and information collected through market and technical research
2. **Select the best idea**
   - Ideas are compared, with some being dropped and others chosen for further research
   - This decision relates closely to **costs** and likely **demand**
   - Research includes looking into forecast sales, size of market share and cost-benefit analysis for each product idea
3. **Develop a prototype**
   - This allows the operations department to see how the product can be manufactured, any problems or difficulties arising from its production and how to fix them
   - Computer simulations are often used to produce 3D prototypes on screen
4. **Test launch**
   - The developed product is sold on a small scale to a limited market to see how well it sells before its full launch
   - Changes may be needed prior to an expensive, large-scale launch
   - Digital products such as apps and software run beta versions, which is a method of test-launching
5. **Full launch of the product**
   - The finalised version of the product is launched to the entire target market

Sources of ideas for product development

- Product development is important for helping a business stay competitive, grow and meet changing customer needs
- Before creating a new product, a business will explore different sources to get ideas and understand what customers might want

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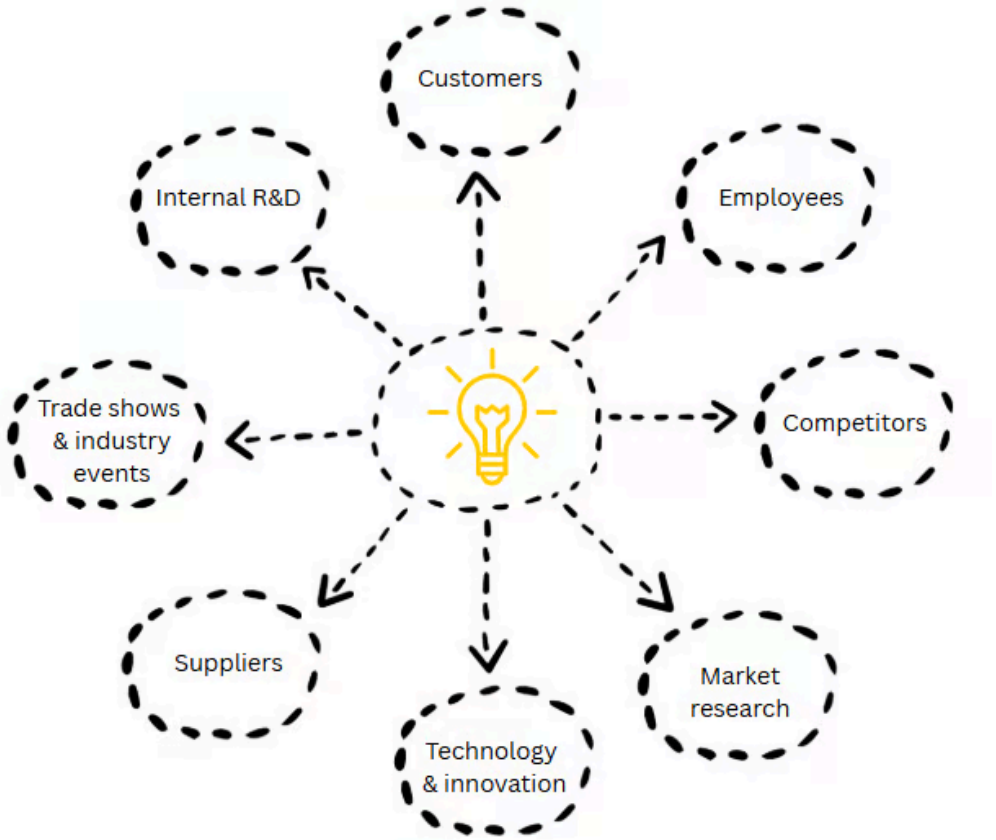

Ideas can be generated by employees, from customer feedback and through market research

- **Customers**
  - Feedback, complaints, reviews or suggestions can highlight what people want or need
- **Employees**
  - Staff working in sales, customer service or production often notice problems or spot opportunities for new ideas
- **Competitors**
  - Looking at what rivals are offering can inspire new ideas or show gaps in the market
- **Market research**
  - Surveys, focus groups and trends analysis help businesses understand customer needs and spot new opportunities
- **Technology and innovation**

- New inventions or improvements in technology can lead to better or completely new products
- **Suppliers**
  - A supplier might offer new materials or production methods that help create a new product
- **Trade shows and industry events**
  - These can show off the latest trends and generate ideas for new products
- **Internal research and development (R&D)**
  - Some businesses invest in their own R&D teams to create new products or improve existing ones

The importance of research and development

- **Product development** is a key part of business success, as it supports growth and helps a business compete and meet customer needs

Key benefits of product development

- **It helps a business stay competitive**
  - Developing new products allows a business to keep up with changes in customer preferences and technology
    - E.g. **Apple** regularly updates its iPhone models to stay ahead of rival smartphone brands
- **It supports business growth**
  - New products can open up new markets or increase sales, helping the business grow in size and profits
    - E.g. **Greggs** launched vegan options to attract new customers and grow its market share
- **It builds customer loyalty and strengthens the brand**
  - Offering fresh and useful products shows customers that the business listens and adapts to their needs
    - E.g. **Coca-Cola** released zero-sugar versions of its drinks to meet health concerns and keep loyal customers

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The need to forecast sales

- Sales forecasts **predict future revenues** based on past sales figures
  - They are an important **business planning tool**
  - They can be used to **identify trends in sales** which can then be compared with the market as a whole
- They commonly focus on what will happen **in the future** to:
  - The volume and value of sales
  - The size of the market
  - Sales as a result of promotional activity
  - Sales as a result of cyclical factors
- Sales forecasts are an important tool to **support planning** and improve the **validity** of **cash flow forecasts**
- Businesses use sales forecasts to **determine resource requirements**, such as
  - How many staff are needed
  - How much stock is required
  - Whether **capacity** needs to be expanded or reduced
  - Whether equipment needs to be upgraded or replaced
  - How much and **which type of finance** is required
  - Whether and when **promotional activity** (e.g. advertising) is required

Time series analysis

- **Raw sales data** on its own may not be very useful for sales forecasting
  - It can be hard to spot patterns, trends or seasonal changes by looking at rows of data or, even, data in a graph, especially if sales are volatile
    - In the example below, it would be **difficult to draw a line of best fit** in order to extrapolate and make sales forecasts

Example of raw data plotted in a chart

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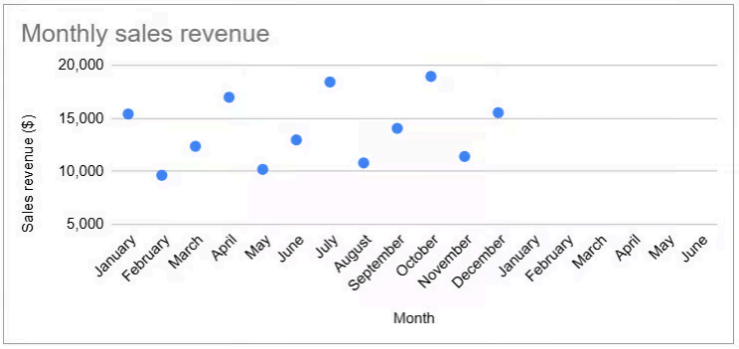
Month	Sales revenue ($)
January	15,425
February	9,647
March	12,379
April	17,002
May	10,204
June	12,985
July	18,441
August	10,813
September	14,070
October	18,965
November	11,420
December	15,550



Sometimes it is difficult to identify a line of best fit to extrapolate a trend as raw sales data is volatile

- Businesses may uses **time series analysis** to smooth data, making it easier to predict likely sales in the future based on past performance

How to conduct a time series analysis

Step 1: Calculate moving totals

- In this case, a **three-month moving average** will be calculated
- Add together sales for January, February and March
- Place the total alongside March

$$= 15{,}425 + 9{,}647 + 12{,}379$$

$$= 37{,}451$$

- Now do the same for February, March and April
- Place the total alongside March

$$= 9{,}647 + 12{,}379 + 17{,}002$$

$$= 39{,}028$$

- Now do the same for March, April and May
- Place the total alongside May

$$= 12{,}379 + 17{,}002 + 10{,}204$$

$$= 39.585$$

- Repeat the process for each subsequent group of three months

Completed table: 3-month moving totals

Your notes

Month	Sales revenue ($)	3-month moving total
January	15,425	
February	9,647	
March	12,379	37,451
April	17,002	39,028
May	10,204	39,585
June	12,985	40,191
July	18,441	41,630
August	10,813	42,239
September	14,070	43,324
October	18,965	43,848
November	11,420	44,455
December	15,550	45,935

Add sales for **January, February** and **March**
= 15,425 + 9,647 + 12,379
= **37,451**

Place this total alongside **March**

Then add sales for **February, March** and **April**
= **39,028**

Place this total alongside **April**

Then add sales for **March, April** and **May**
= **39,585**

Place this total alongside **May**

Repeat for each group of three months

The three month moving total is a sum of each three months of sales data

Your notes

Step 2: Calculate moving averages

- Divide March's moving total by 3
- Place the average alongside February

$$= 37{,}451 \div 3$$

$$= 12{,}484$$

- Divide April's moving total by 3
- Place the average alongside March

$$= 39{,}028 \div 3$$

$$= 13{,}009$$

- Divide May's moving total by 3
- Place the average alongside April

$$= 39{,}585 \div 3$$

$$= 13{,}195$$

- Repeat the process for each subsequent group of three months

Completed table: 3-month moving averages

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Month	Sales revenue ($)	3-month moving total	3-month moving average
January	15,425		
February	9,647		12,484
March	12,379	37,451	13,009
April	17,002	39,028	13,195
May	10,204	39,585	13,397
June	12,985	40,191	13,877
July	18,441	41,630	14,080
August	10,813	42,239	14,441
September	14,070	43,324	14,616
October	18,965	43,848	14,818
November	11,420	44,455	15,312
December	15,550	45,935	

Divide **March** 3-month moving total by 3
= 37,451 ÷ 3
= **12,484**

Place this average alongside **February**

Then divide **April** 3-month moving total by 3
= **13,009**

Place this total alongside **March**

Repeat for each month - except the first and last

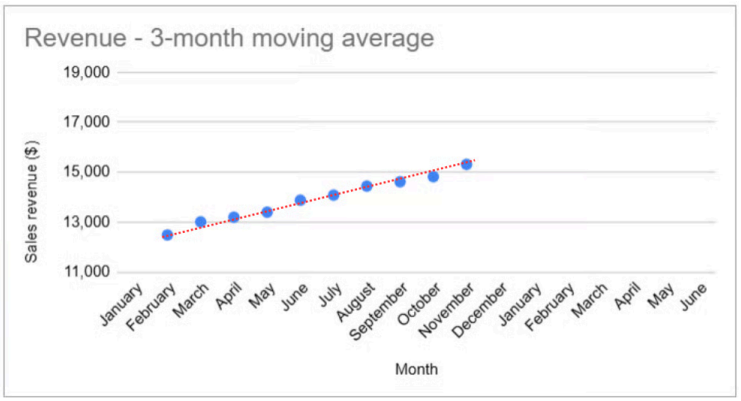
The moving average smooths data that can be plotted in a graph to allow a line of best fit to be applied

Step 3: Plot the moving averages and identify the line of best fit

- Include as many data points above the line as below the line.
  - Watch out for outlying data - if there is more than one outlier above the line, adjust your line of best fit upwards
  - Similarly, if there is more than one outlier below the line, adjust your line of best fit downwards

The line of best fit

Month	3-month moving average
January	
February	12,484
March	13,009
April	13,195
May	13,397
June	13,877
July	14,080
August	14,441
September	14,616
October	14,818
November	15,312
December	



Line of best fit

With smoothed data, a line of best fit can be identified

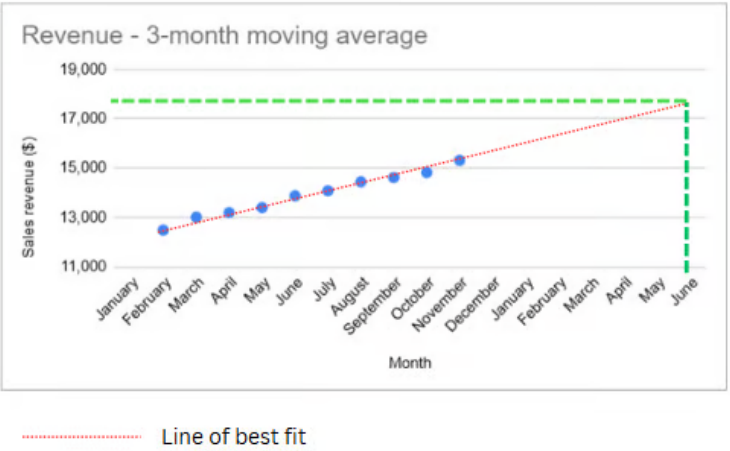
Step 4: Extrapolate to forecast future sales

- Extend the line of best fit to predict future sales values

Your notes

Extension of the line of best fit

Month	3-month moving average
January	
February	12,484
March	13,009
April	13,195
May	13,397
June	13,877
July	14,080
August	14,441
September	14,616
October	14,818
November	15,312
December	



Sales of 17,750 in June can now be forecast by extending the line of best fit

- In this cases, the line of best fit is extended to June, allowing a prediction of 17,750 sales in that month

Qualitative sales forecasting

- **Qualitative sales forecasting** involves predicting future sales based on **non-numerical data**
  - This may include **expert opinions**, market experience or customer feedback, rather than historical figures or statistical analysis
- Qualitative forecasting relies on **judgement, knowledge** and **experience** rather than data modelling
- Methods include
  - **Expert opinion**
    - Insights from experienced managers, industry specialists or consultants
  - **Sales force estimates**
    - Forecasts made by sales staff based on their direct interaction with customers
  - **Customer surveys**
    - Using feedback from potential buyers to estimate future demand
- It is especially useful when a business
  - is launching a **new product** with no past sales data
  - is entering a **new market** or responding to major market changes
  - wants to understand **consumer attitudes** and future buying behaviour

Advantages of qualitative sales forecasting

Your notes

- **Useful when historical data is unavailable**
  - This makes it ideal for use with new products or markets
- **Incorporates expert judgement**
  - Makes use of real-world knowledge and insights that numbers can't always capture
- **Flexible and adaptable**
  - Can be quickly adjusted based on changing market conditions
- **Encourages team involvement**
  - Sales teams and managers contribute ideas, improving internal communication and morale

Your notes

Disadvantages of qualitative sales forecasting:

- **Subjective and prone to bias**
  - Forecasts may reflect opinions rather than facts
- **Less accurate than quantitative methods**
  - Without numerical data, predictions may be vague or inconsistent
- **Hard to measure reliability**
  - It is difficult to test or prove whether a forecast is correct until actual sales occur
- **Time-consuming**
  - Gathering expert opinions or conducting surveys takes time and requires coordination

The impact of sales forecasting on business decisions

- Accurate sales forecasting can help a business **plan ahead** and make informed decisions across functional areas
- However, sales factors should be considered alongside other research on likely future business performance

Sales forecasting and functional decision making

Business function	Impact of sales forecasting
Marketing	▪ **Promotional planning** ▪ Helps identify when to increase or reduce advertising ▪ E.g. If sales are expected to fall in summer, promotions may be introduced to boost demand

	▪ **Target setting**
	  ▪ Allows marketing teams to set realistic sales goals and measure performance
	  ▪ E.g. A retailer may forecast higher sales in December and increase ad spend before Christmas

Operations	▪ **Resource planning**
	  ▪ Ensures the right amount is produced to meet expected demand
	  ▪ E.g. A food company forecasts demand for a festival and increases production in advance.
	▪ **Stock control**
	  ▪ Helps prevent overproduction or shortages
	  ▪ E.g. An electronics firm reduces production during slow periods to avoid excess stock

Human Resources	▪ **Staffing needs**
	  ▪ Helps plan recruitment, training, and staff scheduling
	  ▪ E.g. A hotel hires extra staff for a predicted rise in summer guests
	▪ **Avoids overstaffing**
	  ▪ Saves costs when lower demand is forecasted
	  ▪ E.g. A retail store reduces staff hours in January when sales are expected to drop

Finance	▪ **Budgeting and cash flow**
	  ▪ Improves financial planning and helps manage income and spending
	  ▪ E.g. A business delays purchases if a fall in sales is forecasted
	▪ **Investment planning**
	  ▪ Encourages or delays investment based on expected revenue
	  ▪ E.g. A company forecasting growth may invest in expanding to new locations

Your notes